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CODE OF PROJECT :

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-- PROJECT: UNIVERSITY BADMINTON DBMS (COMPLETE UNIFIED WORKSPACE)  
-- INSTRUCTIONS: Copy and paste this entire code block into your SQL  
tab.  
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-- STEP 1: CLEAN CONFLICTING OBJECTS (ORDER MATTERS DUE TO FOREIGN  
KEYS)  
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```

```
--  
DROP TABLE IF EXISTS Scores;  
DROP TABLE IF EXISTS Matches;  
DROP TABLE IF EXISTS TournamentFixtures;
```

```
DROP TABLE IF EXISTS TeamMembers;
DROP TABLE IF EXISTS Teams;
DROP TABLE IF EXISTS Coaches;
DROP TABLE IF EXISTS Players;
```

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```
-- STEP 2: CREATE BADMINTON MANAGEMENT RELATIONAL SCHEMA
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```
CREATE TABLE Players (
    player_id INT AUTO_INCREMENT PRIMARY KEY,
    student_id VARCHAR(50) UNIQUE NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    department VARCHAR(100) NOT NULL,
    semester INT NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    contact_no VARCHAR(20) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE Coaches (
    coach_id INT AUTO_INCREMENT PRIMARY KEY,
    coach_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    specialization VARCHAR(100) DEFAULT 'General Badminton',
    experience_years INT NOT NULL
);
```

```
CREATE TABLE Teams (
    team_id INT AUTO_INCREMENT PRIMARY KEY,
    team_name VARCHAR(100) UNIQUE NOT NULL,
    captain_id INT,
    coach_id INT,
    FOREIGN KEY (captain_id) REFERENCES Players(player_id) ON DELETE
SET NULL,
    FOREIGN KEY (coach_id) REFERENCES Coaches(coach_id) ON DELETE SET
NULL
);
```

```
CREATE TABLE TeamMembers (
```

```

    team_id INT,
    player_id INT,
    PRIMARY KEY (team_id, player_id),
    FOREIGN KEY (team_id) REFERENCES Teams(team_id) ON DELETE CASCADE,
    FOREIGN KEY (player_id) REFERENCES Players(player_id) ON DELETE
CASCADE
);

```

```

CREATE TABLE TournamentFixtures (
    tournament_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(150) NOT NULL,
    category ENUM('Singles', 'Doubles', 'Mixed Doubles') NOT NULL,
    start_date DATE NOT NULL,
    venue_court VARCHAR(50) NOT NULL,
    status ENUM('Upcoming', 'Ongoing', 'Completed') DEFAULT 'Upcoming'
);

```

```

CREATE TABLE Matches (
    match_id INT AUTO_INCREMENT PRIMARY KEY,
    tournament_id INT NOT NULL,
    match_datetime DATETIME NOT NULL,
    court_number VARCHAR(20) NOT NULL,
    player1_id INT DEFAULT NULL,
    player2_id INT DEFAULT NULL,
    team1_id INT DEFAULT NULL,
    team2_id INT DEFAULT NULL,
    FOREIGN KEY (tournament_id) REFERENCES
TournamentFixtures(tournament_id) ON DELETE CASCADE,
    FOREIGN KEY (player1_id) REFERENCES Players(player_id) ON DELETE
SET NULL,
    FOREIGN KEY (player2_id) REFERENCES Players(player_id) ON DELETE
SET NULL,
    FOREIGN KEY (team1_id) REFERENCES Teams(team_id) ON DELETE SET
NULL,
    FOREIGN KEY (team2_id) REFERENCES Teams(team_id) ON DELETE SET NULL
);

```

```

CREATE TABLE Scores (
    score_id INT AUTO_INCREMENT PRIMARY KEY,
    match_id INT NOT NULL,
    set_number INT NOT NULL,
    side1_score INT NOT NULL,
    side2_score INT NOT NULL,
    set_winner_name VARCHAR(100) DEFAULT 'TBD',

```

```

        FOREIGN KEY (match_id) REFERENCES Matches(match_id) ON DELETE
CASCADE
);

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-- STEP 3: SEED PRODUCTION DATA FOR TESTING RUNS
-- Use LAST_INSERT_ID() or omit PK to avoid ID mismatch
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-- Seed Players
INSERT INTO Players (student_id, full_name, department, semester,
email, contact_no) VALUES
('CS-001', 'Zain Malik', 'Computer Science', 4, 'zain.malik@uni.edu',
'03001234567'),
('CS-002', 'Ayesha Khan', 'Computer Science', 4, 'ayesha.k@uni.edu',
'03007654321'),
('EE-001', 'Hamza Ali', 'Electrical Engineering', 6, 'hamza.a@uni.edu',
'03219876543'),
('BA-001', 'Nimra Ahmed', 'Business Administration', 2,
'nimra.a@uni.edu', '03331122334');

-- Seed Coaches
INSERT INTO Coaches (coach_name, email, specialization,
experience_years) VALUES
('Coach Tariq', 'tariq.badminton@uni.edu', 'Advanced Smash Metrics',
8),
('Coach Sofia', 'sofia.sports@uni.edu', 'Footwork & Agility', 5);

-- Seed Teams & Assign Captains/Coaches using LAST_INSERT_ID
INSERT INTO Teams (team_name, captain_id, coach_id) VALUES
('CS Smashers', 1, 1),
('EE Warriors', 3, 2);

-- Assign Players to Teams
INSERT INTO TeamMembers (team_id, player_id) VALUES
(1, 1), (1, 2),
(2, 3), (2, 4);

-- Seed a Tournament Fixture - omit tournament_id

```

```

INSERT INTO TournamentFixtures (title, category, start_date,
venue_court, status) VALUES
('Inter-Departmental Badminton Championship 2026', 'Singles', '2026-11-
20', 'Main Indoor Hall', 'Upcoming');

-- Seed a Match - omit match_id, use LAST_INSERT_ID for tournament_id
INSERT INTO Matches (tournament_id, match_datetime, court_number,
player1_id, player2_id) VALUES
(LAST_INSERT_ID(), '2026-11-20 10:00:00', 'Court 1', 1, 3);

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-- STEP 4: RUN TIME SEPARATE FEATURE EXECUTIONS
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-- Feature 1: Player Registration Lookup
SELECT student_id, full_name, department, semester, email
FROM Players
WHERE department = 'Computer Science';

-- Feature 2: Update Player Contact Details
UPDATE Players
SET contact_no = '03210001122'
WHERE student_id = 'CS-001';

-- Feature 3: Read Team Layouts with assigned Coach Profiles
SELECT t.team_name, p.full_name AS captain_name, c.coach_name
FROM Teams t
LEFT JOIN Players p ON t.captain_id = p.player_id
LEFT JOIN Coaches c ON t.coach_id = c.coach_id;

-- Feature 4: View Team Members Allocation Directory
SELECT t.team_name, p.full_name AS member_name, p.department
FROM TeamMembers tm
JOIN Teams t ON tm.team_id = t.team_id
JOIN Players p ON tm.player_id = p.player_id;

-- Feature 5: Create/Monitor Tournament Fixtures States
SELECT title, category, venue_court, status
FROM TournamentFixtures
WHERE status = 'Upcoming';

```

-- Feature 6: Schedule Matches Verification Loop

```
SELECT m.match_id, t.title AS tournament, m.match_datetime,  
m.court_number,  
p1.full_name AS player_1, p2.full_name AS player_2  
FROM Matches m  
JOIN TournamentFixtures t ON m.tournament_id = t.tournament_id  
LEFT JOIN Players p1 ON m.player1_id = p1.player_id  
LEFT JOIN Players p2 ON m.player2_id = p2.player_id;
```

-- Feature 7: Live Match Score Entry Execution

-- Use subquery to get match_id instead of hardcoding 50

```
INSERT INTO Scores (match_id, set_number, side1_score, side2_score,  
set_winner_name)  
SELECT match_id, 1, 21, 19, 'Zain Malik' FROM Matches WHERE  
player1_id=1 AND player2_id=3  
UNION ALL  
SELECT match_id, 2, 18, 21, 'Hamza Ali' FROM Matches WHERE player1_id=1  
AND player2_id=3;
```

-- Feature 8: Automatic Calculation & Score History Lookups

```
SELECT s.set_number, s.side1_score, s.side2_score, s.set_winner_name  
FROM Scores s  
JOIN Matches m ON s.match_id = m.match_id  
WHERE m.player1_id=1 AND m.player2_id=3  
ORDER BY s.set_number ASC;
```

-- Feature 9: Track Coach Assignments Map

```
SELECT coach_name, specialization, experience_years  
FROM Coaches  
WHERE experience_years >= 5;
```

-- Feature 10: Cascading Clean Up Simulation Validation

-- Delete by student_id to avoid hardcoding PK

```
DELETE FROM Players WHERE student_id = 'BA-001';
```

OUTPUT:

Output:

student_id	full_name	department	semester	email
CS-001	Zain Malik	Computer Science	4	zain.malik@uni.edu
CS-002	Ayesha Khan	Computer Science	4	ayesha.k@uni.edu

team_name	captain_name	coach_name
CS Smashers	Zain Malik	Coach Tariq
EE Warriors	Hamza Ali	Coach Sofia

team_name	member_name	department
CS Smashers	Zain Malik	Computer Science
CS Smashers	Ayesha Khan	Computer Science
EE Warriors	Hamza Ali	Electrical Engineering
EE Warriors	Nimra Ahmed	Business Administration

title	category	venue_court	status
Inter-Departmental Badminton Championship 2026	Singles	Main Indoor Hall	Upcoming

match_id	tournament	match_datetime	court_number	player_1	player_2
1	Inter-Departmental Badminton Championship 2026	2026-11-20 10:00:00	Court 1	Zain Malik	Hamza Ali

set_number	side1_score	side2_score	set_winner_name
1	21	19	Zain Malik
2	18	21	Hamza Ali

coach_name	specialization	experience_years
Coach Tariq	Advanced Smash Metrics	8
Coach Sofia	Footwork & Agility	5